4748-120 (E-volve) and 4748-220 (Paper-based) Functional Skills Mathematics Level 2

Chief Examiner Report

Version 1.6
Last modified 3-September-2025
For external use

Document revision history

Version	Changed by	Summary of change	Approval date
1.3	Assessment	New Template/updated	09/02/2023
1.4	Assessment	New version	21/11/2023
1.5	Assessment	Updated version	04/09/2024
1.6	Assessment	Updated version	03/09/2025

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1. Introduction

The purpose of this document is to provide centres with feedback on the performance of candidates for 4748-120 and 4748-220 Functional Skills Mathematics Level 2.

Centres should read this report in conjunction with the Guidance for Delivery document.

2. Overall Performance

This report covers the period from September 2024 to August 2025.

Candidates are expected to adequately cover underpinning knowledge (UPK) and problem solving (PS) skills within the assessment, including completing part of the assessment without a calculator. Candidates are therefore required to undertake a two-part assessment.

	Part 1 **Calculator not permitted** **(25 minutes)**	**Part 2** **Calculator permitted** **(1 hour 20 minutes)**
Underpinning knowledge **(15 marks = 25%)**	10 single mark context free questions	5 single mark context free questions
Problem solving **(45 marks = 75%)**	2 problem solving questions with practical context (total 5 marks)	1 single mark check (for sense of given result) 9 problem solving questions with practical context (mark tariff between 2 and 6 marks each, total 39 marks)
Overall total marks 60	Total 15 marks	Total 45 marks

While many candidates coped well with the assessment requirements and have achieved pass grades; a significant percentage of candidates performed less well. Approximately 30% of candidates achieved 20 marks or less and 10% less than 10 marks. As an example, some candidates were unable to complete basic calculations involving fractions and/or percentages and did not attempt the problem-solving questions in Part 2 of the assessment.

2.1. Areas of good performance

A large number of candidates produced well worked solutions to the problems set.

Many candidates coped well with both the isolated, context free questions (Underpinning Knowledge) and used their knowledge and skills to recognise and obtain a solution or solutions to a number of complex problems (Problem Solving).

Many candidates coped with calculation requirements and understood the principles of basic operations (addition, subtraction, multiplication, division and BIDMAS) and dealt with fractions, decimals, percentages, ratios/proportion and scaling. Basic geometry and algebra were dealt with well by successful candidates.

Understanding and interpretation of scales, scale plans and various data presentations were dealt with satisfactorily.

Many candidates dealt with statistical problems competently, chose and accurately calculated appropriate averages, including estimation from grouped data, and ranges. They also dealt competently with two event probability with tabulated data and probability trees.

Successful candidates gave clear explanations of their results and demonstrated understanding of the problem contexts.

2.2. Areas for development

General observations

Script marking shows that some candidates were not able to meet the demands of the Level 2 papers. Some candidates could not make accurate calculations using fractions, percentages and ratios; round to given numbers of decimal places; read and apply scales; group data and make statistical calculations including probability.

Candidates are expected to show their working in order to be eligible for compensation marks in cases where they have not achieved a fully correct answer. This has been a particularly important issue for some online candidates who use pencil and paper for initial workings out and neglect to transfer some or all of their working to the online script.

Some online candidates did not appear to have had sufficient practice in using the diagram or chart tools and therefore lost a significant number.

2.3. Underpinning knowledge (UPK) questions

There are 15 underpinning knowledge questions, generally with no contextual setting. Ten of these questions are in the ‘calculator not permitted’ section. In this section candidates have just under two minutes per question and therefore need to remain aware of the time limitations.

Many of the questions are calculations involving an understanding of fractions, percentages and ratios. Some involve basic geometry (calculation of angles); calculations of areas, perimeters and volumes; and statistics questions including probability.

Many of the difficulties encountered by candidates in the UPK sections translate into difficulties in dealing with problem solving (PS) questions, where the same issues arise for solving questions in more complex contexts.

Areas where some candidates performed less well –

Under Pinning Knowledge questions

The following under pinning knowledge question types were the ones that were less successfully attempted by more than half the candidates over last year's assessments.

Underpinning knowledge	Examples/explanation
SCS22 Calculate values of angles	Calculation of angle (isosceles triangle)
	Calculation of angle (series of isosceles triangles)
	Calculation of angle (polygon using given formula)
SCS11 Understand and calculate using ratio	Use of scaling ratio (map scale given, conversion of map measurement to actual distance)
SCS6 Calculate percentage change	Calculation of percentage decrease (viewing figures from one period to next)
	Percentage increase (number of visitors year to next year)
SCS14 Convert between imperial and metric units	Speed conversion calculation (conversion mph > kph)
SCS12 Follow the order of precedence of operators including indices	BIDMAS calculation (common error, omission of minus symbol for negative number)
SCS10 Multiply decimals up to 3 dp	Multiplication of decimals – non calculator (difficulties with position of decimal point
SCS27 Express probabilities as fractions, decimals or percentages	Probability calculation (win / loss game statistics)
SCS23 calculate the mode of a set of quantities	Identification of modal value from tabulated data

SCS4 Know the equivalence between fractions and decimals	Conversion of improper fraction to decimal fraction
SCS14 Convert between metric and imperial units of length	Conversion of inches to cm (given conversion factor)
SCS5 Work out percentages of amounts	Calculation of percentage of decimal fraction of one million eg 6% of 0.4million

2.4. Problem solving (PS) questions

There are 12 problem solving questions, including two in the 'calculator not permitted' section.

Many candidates found difficulty coping with the style and scope of problem solving questions, given the number of different contexts presented.

There is little or no scaffolding within a question and little guidance given beyond a start point and a finish point. Candidates are expected to choose an appropriate approach and methods as well as carry out calculations. They will also be given opportunities to interpret information and are expected to provide explanations relating their results to the question context.

Areas where some candidates performed less well - Problem solving

Units

Misunderstanding units, particularly relating to linear dimensions (mm, cm, m and km) and those of time, prevented some candidates from successfully completing questions. Some candidates confuse linear and area measurement, equating both.

Many candidates did not show units either in their answers or workings. Although candidates will not be repeatedly penalised for this, the absence of units can lead to confusion for them as their answer develops.

Many candidates ignored the need to make use of the £ sign and some gave answers in incorrect money format,

e.g. an answer £107.30 written as £107.3 will be penalised.

Explanations

Problem solving questions may specify a requirement for explanation, sometimes asked for as ‘comments’. Explanations of results in context have been a problem for many candidates as they missed an extra mark in questions that required ‘rounding off’ with a brief summary of how their results answered the problem outlined in the question.

The explanation should link with (refer back to) the problem stated in the question using values calculated.

Example of a problem	Explanation/Example of an expected answer
A company claims that the area rented for an exhibition last year was 4% smaller than the area on offer this year	***Result of calculation*:** area last year is $5m^2$, this year it is $4.8m^2$ ***Answer*:** the company is correct ***Explanation:*** last year is $0.2m^2$ = 4% smaller this year
A student wants the best deal for travel to and from a concert.	***Result of calculation*:** return ticket costs £37, two single tickets cost £16 each ***Answer*:** the student should buy two single tickets ***Comment:*** two single tickets are £5 cheaper
Questions involving graphical interpretation	Many candidates lost marks by omitting axis labels, failing to draw lines of best fit accurately (or at all) and drawing accurately using a straight edge (paperbased) or the straight line function (online).
	Many candidates were unable to extrapolate given graphs, and many did not read accurately from line graphs. e.g. some candidates mixed up readings by misunderstanding the axes
	Explanations from lines of best fit often missed reference to trends (many candidates pick out specific plots rather than trends).
1 single mark check (for sense of given result)	Some candidates struggle with this type of question where there is an expectation that they will consider whether a result is sensible or not. There is usually no requirement to make a calculation (although this is optional) and

	candidates should be advised to look at given data or graphical evidence relating to a given comment or result. e.g. 0.6 x 0.9 cannot be >1 e.g the average return journey speed of a school bus will not be 90mph e.g. graphical information can be misleading and misinterpreted if scales are different, axes are unlabelled or units are missing

Check for sense: Many candidates struggled with this type of question where there is an expectation that they will consider whether a result is sensible or not. Candidates should be advised to look at given data or graphical evidence relating to a given comment or result.

They should be aware that this question will not necessarily need a calculation.

The following problem-solving question types from last year's assessments were not successfully achieved by more than half the candidates.

Problem-solving questions	Example
SCS15 Calculate using compound measures	Determination of arrival time from speed/distance /time calculation
	Calculation of average speed for two journeys
SCS25 Use the mode to compare two sets of data	Identification of modal values from tabulated data and comparison of results
SCS6 Calculate percentage change	Calculation of scaled size for model given scale diagram and percentage change from results
SCS23 Calculate the median of a set of quantities	Identification and calculation of suitable average for given set of wages (candidates expected to choose median as data has outlier)
SCS17 Use formulae to calculate volumes and surface areas of 3-D objects	Calculation of surface area of pillar and amount of paint required (given coverage). Formula given for surface area of sphere.

SCS26 Work out the probability of combined events including the use of diagrams	Calculation of volume of ramp (formula given) and cost of construction materials
	Calculation of two event probability (weather – two consecutive days)
	Calculations from interpretation of probability tree diagram
SCS13 Calculate amounts of money including percentage decreases	Calculation of original cost of item with VAT added
	Calculations to check validity of one third off price offer
SCS18 /16 Calculate actual dimensions from scale diagrams / calculate areas of 2-D shapes including no n rectangular composite shapes	Calculation of complex area from scale diagram and amount of material required to cover area (given rate of coverage)
SCS28 Interpret scatter diagrams and recognise positive and negative correlation	Interpretation of scatter graph (recognition of outlier and reading from trendline)

3. Recommendations/Advice for centres

The assessment, is based on the DfE Subject content functional skills Maths

The assessment is based on the 28 Subject Content Statements (SCS), and the general descriptions preceding each section. The content at Level 2 subsumes and builds upon the content at all the lower levels (ie Entry 1 to Level 1). The Level 2 Functional Skills Mathematics is considered to be the equivalent of GCSE grade 4/5.

Centres should consider carefully whether a candidate is operating at an appropriate level for entry at Level 2.

There are two platforms, paper-based and online available for this assessment. Centres should ensure that an appropriate choice of platform is made for candidates based on each candidate's need and preference.

Centres should advise candidates about appropriate 'exam technique' particularly with regard to attempting questions in order. Candidates may attempt questions in any order.

Candidates who choose to access assessment online need to be prepared not only in terms of the prescribed Functional Skills Standards, but also in terms of using the E-volve platform. They must be well practised in the use of the presentation tools (tables, diagrams, charts and graphs). The html Maths Familiarisation Test, available on Open Assess, enables candidates to practise online tools available.

Online candidates must also understand how to insert sufficient text, e.g. to show calculations and working, so that potential compensation marks, in the event of incorrect answers, are accessible. The importance of showing working in paper-based assessment should be stressed for the same reason.

4. Additional Information

Centres should be aware that pass marks may vary from paper to paper as a result of an awarding process undertaken by City & Guilds. Any difference in pass marks reflects the perceived and actual difference in demand of the exam papers and the questions themselves. Therefore, it is possible that two candidates with the same score may have different overall results (pass or fail) if they sat different papers.

Centres are reminded of the Indicative Pass Mark document on the City & Guilds website.

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City & Guilds
5-6 Giltspur Street
London EC1A 9DE